Tangent vectors

For S in $\mathbb{R}^3$ we can define tangent vectors using velocities of curves

γ smooth $I \to S$, $\gamma(0) = p \in S \Rightarrow \dot{\gamma}(0)$ is tangent to S at p

For manifolds, we will not have an ambient space to take derivatives in

$\Rightarrow$ associate tangent vectors to directional derivatives

define: A <u>tangent vector</u> to S at p is a linear map $X : C^\infty(S) \to \mathbb{R}$ such that

1) If $f = g$ in a nbhd of p, $Xf = Xg$ (local at p)
2) $X(fg) = g(p) \cdot Xf + f(p) \cdot Xg$ ("derivation")

$T_pS := \{\text{tangent vectors to } S \text{ at } p\}$

examples: 1) velocity of a curve $\gamma : I \to S$

$$\dot{\gamma}(t) : f \longmapsto \frac{d}{dt} f(\gamma(t))$$

This gives $\dot{\gamma}(t) \in T_{\gamma(t)}S$ for each t

2) coordinate vectors: let $\sigma : U \to S$ be a regular patch

$f \in C^\infty(S)$ can be written as $f(u^1, u^2)$ (technically this is $f \circ \sigma$)

$\Rightarrow$ define $\partial_i := \frac{\partial f}{\partial u^i}(u^1, u^2) \in T_pS$ where $p = \sigma(u^1, u^2)$

In the coordinate patch we can take u^1, u^2 as locally defined functions

Given $X \in T_pS$, define coefficients $a^j := Xu^j$

Lemma In a local patch, a tangent vector $X \in T_pS$ can be written as

$$X = a^1 \partial_1 + a^2 \partial_2$$

(T_pS is 2-dimensional and coordinates define a basis)

Einstein convention: pairs of upper/lower indices are summed over

$$a^i \partial_i := a^1 \partial_1 + a^2 \partial_2$$

Proof: First note $X(\text{constant fn.}) = 0$

since $X(1) = X(1 \cdot 1) = 1 \cdot X(1) + 1 \cdot X(1)$ and X is linear

Write $f \in C^\infty$ as $f(u^1, u^2)$ near p and assume p corresponds to $u = 0$

Taylor expansion at p:

$$f(u) = f(0) + \partial_i f(0) u^i + g_i(u) u^i \quad \text{where } g_i(0) = 0$$

Since X kills out constants,

$$Xf = \partial_i f(0) \cdot Xu^i + g_i(0) \cdot Xu^i + Xg_i \cdot u^i|_0$$

$$= a^i \partial_i f(0)$$

For $S \subset \mathbb{R}^3$ we can identify with actual vectors

x, y, z define functions on S by restriction

$$V \in T_pS \longrightarrow (V_x, V_y, V_z) \in \mathbb{R}^3$$

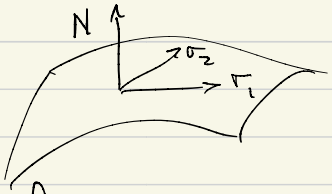
This means $\partial_i \longrightarrow \sigma_i$ in a coord. patch

in $T_pS \longrightarrow$ plane in $\mathbb{R}^3$ with ∂^1, ∂^2 as basis

Normal vector

Given a patch σ, we can define the unit normal

$$N := \frac{\sigma_1 \times \sigma_2}{\|\sigma_1 \times \sigma_2\|}$$



This defines a local coordinate frame

$\{\sigma_1, \sigma_2, N\}$ = basis for $\mathbb{R}^3$ that depends on the surface pt.

def.: A surface is orientable if there is an atlas such that N is consistent across patches

Lengths

For a smooth curve $\gamma : I \to \mathbb{R}^3$,

$$l(\gamma) = \int_I \|\dot{\gamma}\| \, dt$$

On a surface S, if $\gamma : I \to S$ then γ will be tangent at each pt,

def.: the first fundamental form of p is the restriction of the Euclidean inner product to T_pS

$g(A, B) := A \cdot B$ assuming A, B written as vectors in $\mathbb{R}^3$

In coord's, we write g as a 2×2 matrix using the coordinate ∂ basis

$$g_{ij} := g(\partial_i, \partial_j) = \sigma_i \cdot \sigma_j$$

The length integral can be interpreted as $\int ds$ where

$$ds^2 = g_{ij}\, du^i du^j \qquad \text{(in a coordinate patch)}$$

(Given a curve $\gamma(t) = (u^1(t), u^2(t))$,

$$\begin{aligned} ds^2 &= g_{ij}\,(\dot{u}^i\, dt)(\dot{u}^j\, dt) \\ &= g_{ij}\,\dot{u}^i \dot{u}^j\, dt^2 \\ &= \|\dot{\gamma}\|^2 dt^2 \end{aligned}$$

Think of the formula for ds^2 as the legend of the local coord. map

examples:

1) plane $\sigma(x,y) = (x,y,0)$ $\quad \sigma_1 = (1,0,0),\ \sigma_2 = (0,1,0)$

$ds^2 = dx^2 + dy^2$ $\qquad [g_{ij}] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

polar $\sigma(r,\theta) = (r\cos\theta, r\sin\theta, 0)$ $\quad \sigma_1 = (\cos\theta, \sin\theta, 0)$

$\sigma_2 = (r\sin\theta, r\cos\theta, 0)$

$$[g_{ij}] = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}$$

$ds^2 = dr^2 + r^2 d\theta^2$ in polar coord.'s